

9.5 Equivalence Relations

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Definition: An equivalence relation is a relation which is transitive, symmetric, and reflexive.

Definitions: Two elements x and y are equivalent if they are related by an equivalence relation. Denoted $x \sim y$.

Show that $R = \{(a,b) \mid (a,b \in \mathbb{Z}) \wedge (a \equiv b \pmod{m})\}$ is an equivalence relation ($m \in \mathbb{Z}^+$).

Reflexive: $(a,a) \Rightarrow a \equiv a \pmod{m}$

this is always true.

Symmetric: $(a,b) \Rightarrow (b,a)$

If $a \equiv b \pmod{m}$ then $b \equiv a \pmod{m}$


Transitive: If $(a,b) \wedge (b,c)$ then (a,c) .

If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$

then $b = m \cdot x + c$ so

$$a \equiv (mx + c) \pmod{m} \Leftrightarrow a \equiv c \pmod{m}.$$

Definition: Let R be an equivalence relation on a set A . The set of all elements related to $x \in A$ by R is called the equivalence class of x . Denoted $[x]_R$.

If the relation is clear, can omit the R .

For R : "congruent mod 4", what elements are in $[1]$?

$$[1] = \{\dots, -7, -3, 1, 5, 9, \dots\}$$

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$$= \{4n+1 \mid n \in \mathbb{Z}\}$$

Recall the relation

$$R = \{(a,b) \mid (a,b \in \mathbb{R}) \wedge (a \cdot b \geq 0)\}.$$

Is this an equivalence relation?

Yes! We saw it is symmetric,
transitive ; reflexive before!

Find 5 elements in $[-2]$

" " $[\frac{1}{2}]$

" " $[0]$